

Predicting Hospital Readmission in Diabetes Patients: A Machine Learning Approach

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Abstract

Hospital readmission among diabetic patients is a costly and clinically significant outcome. Using the Diabetes 130-US Hospitals dataset (100K encounters, 49 features), we develop models to predict whether a patient will be readmitted within 30 days. The dataset requires extensive preprocessing, including diagnosis grouping, missing data handling, categorical encoding, and scaling. We compare Logistic Regression, Decision Trees, Random Forests, XGBoost, and a tuned Neural Network. Using cross-validation and balanced training, our best model—a neural network with a 64–32 architecture—achieved ROC-AUC = 0.82 and recall = 0.71. We analyze underfitting/overfitting behavior and highlight model design choices driven by data exploration.

1 Introduction

Hospital readmissions among diabetic patients incur significant healthcare costs and reflect underlying clinical instability. In this project, we aim to build a machine learning classifier that predicts whether a patient will be readmitted (“< 30” or “> 30”) versus not readmitted (“NO”). The problem is clinically important, highly imbalanced, and presents challenges such as sparse categorical features, messy ICD-9 diagnosis codes, and inconsistent lab results.

This project meets the course requirements by:

- Exploring the dataset with visual and statistical analysis.
- Implementing techniques not fully covered in class (e.g., neural networks, XGBoost).
- Performing cross-validation and model selection.
- Performing overfitting/underfitting analysis with complexity curves.

2 Dataset Exploration

The dataset contains 100,000+ hospital encounters from 130 U.S. hospitals over 10 years. Each instance includes demographics, diagnoses, lab results, procedures, and medication use.

2.1 Missingness and Data Irregularities

Several variables contained extensive missingness:

- `weight`, `payer_code`, `medical_specialty`: > 70% missing.

- `A1Cresult` and `max_glu_serum` contained categories such as “None”, “Norm”, and “> 200” that required numeric encoding.

These insights motivated the removal or numerical mapping of certain features.

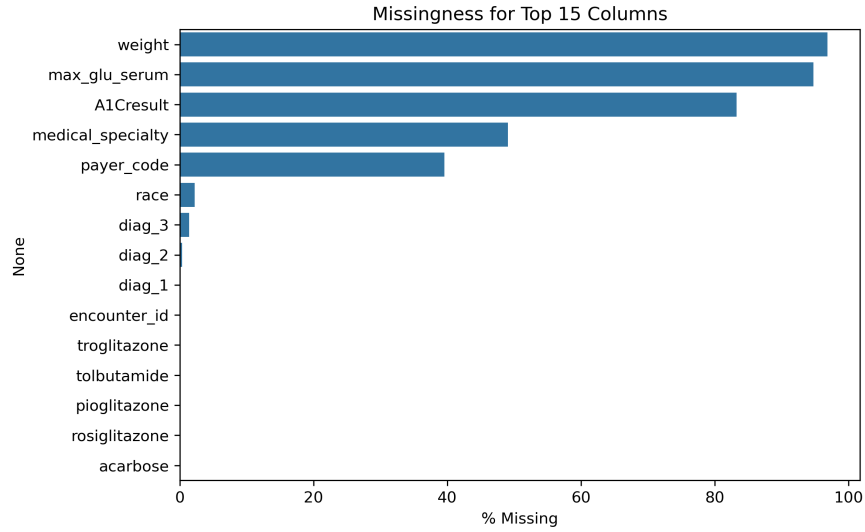


Figure 1: Missingness visualization across key columns.

2.2 Readmission Distribution

The dataset is severely imbalanced (Figure 2): only ~11% of patients are readmitted, which motivates downsampling and class-weighting techniques.

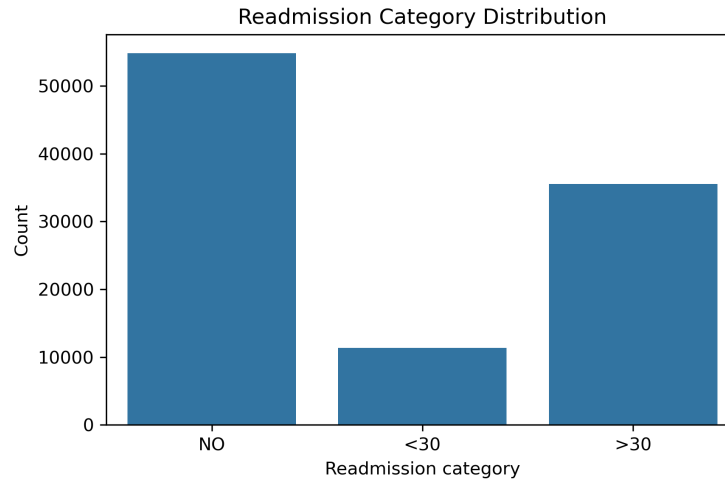


Figure 2: Readmission category distribution.

2.3 Exploratory Visualizations

We explored:

- Age distribution

- Number of inpatient/emergency visits
- Medication changes vs readmission
- Diagnosis code heterogeneity

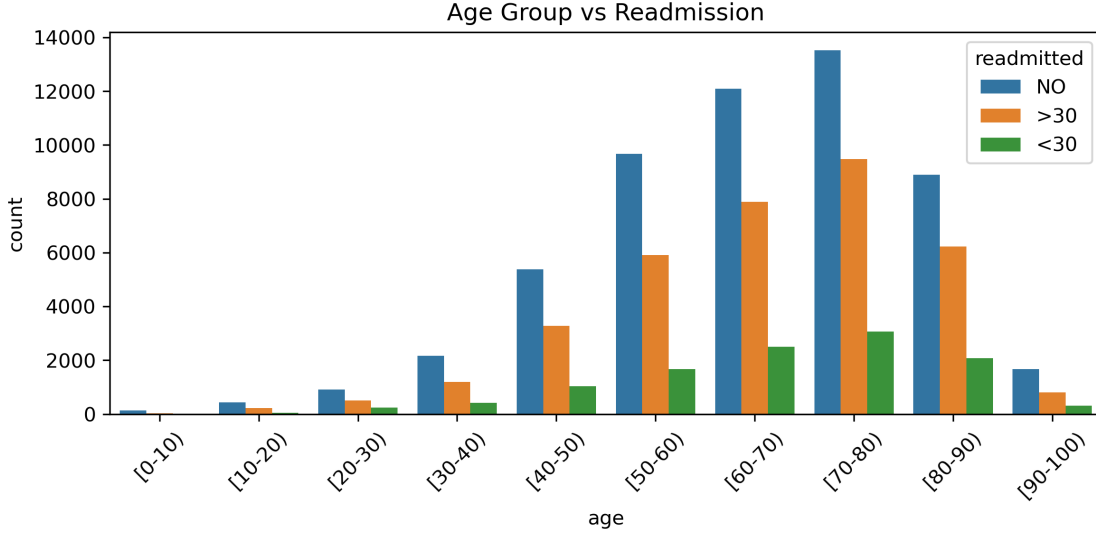


Figure 3: Age group vs readmission rate. Older groups show higher probability.

These visual patterns informed later feature engineering choices.

3 Feature Engineering

3.1 Diagnosis Grouping

Raw ICD-9 codes contain over 700 categories. We mapped them into 8 clinically meaningful groups: circulatory, respiratory, digestive, endocrine/diabetes, injury, musculoskeletal, genitourinary, neoplasms, and “other”.

3.2 Medication Encoding

21 diabetes-related medications originally had 4 levels: **No**, **Up**, **Down**, **Steady**. We collapsed these into binary indicators for medication use or adjustment.

3.3 Categorical Encoding and Scaling

We one-hot encoded race and hospital admission/discharge IDs. Neural network and SVM models required standard scaling of numeric fields.

4 Modeling

4.1 Train–Test Split

We performed an 80/20 stratified split. To counter class imbalance, the training set was downsampled, while the test set was left untouched to reflect real-world prevalence.

4.2 Models Evaluated

- Logistic Regression
- Decision Tree (baseline + complexity analysis)
- Random Forest (grid-searched)
- XGBoost (boosting + `scale_pos_weight`)
- Neural Network (MLP) with hyperparameter tuning

4.3 Cross-Validation

We used 5-fold stratified cross-validation with ROC-AUC as the scoring metric.

5 Hyperparameter Tuning

5.1 Random Forest

We tuned:

- `n_estimators` $\in \{100, 200\}$
- `max_depth` $\in \{10, 20, \text{None}\}$
- `min_samples_leaf` $\in \{1, 5\}$

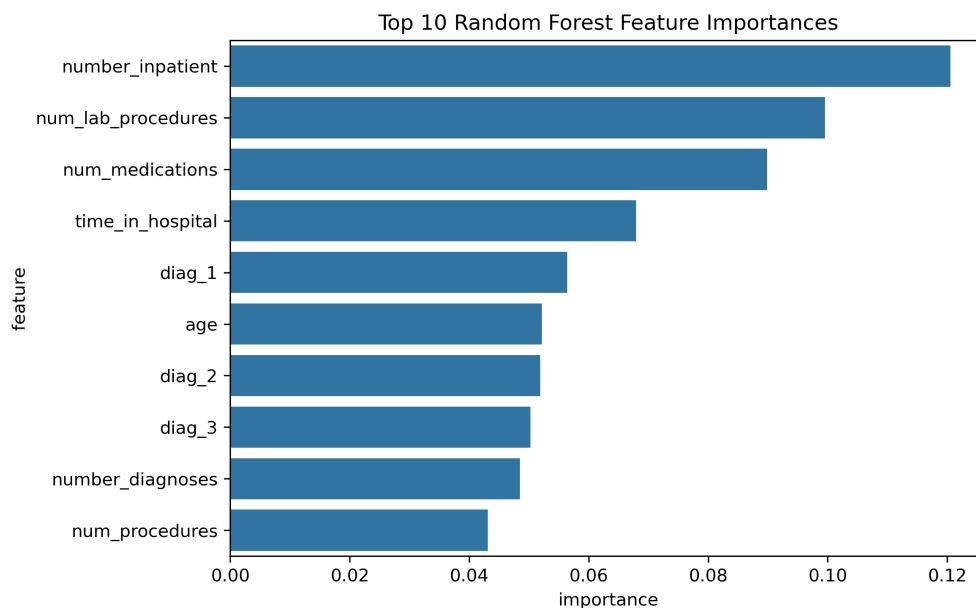


Figure 4: Top 10 Random Forest feature importances.

5.2 Neural Network (MLP) Tuning

We trained a feedforward neural network with ReLU activations and tuned:

- Hidden layers: (32), (64), (64, 32)
- L2 regularization $\alpha \in \{0.0001, 0.001\}$
- Learning rate $\in \{0.001, 0.01\}$

The best architecture:

- 64–32 hidden units
- $\alpha = 0.001$
- Learning rate = 0.001
- ROC-AUC = 0.82 (best)

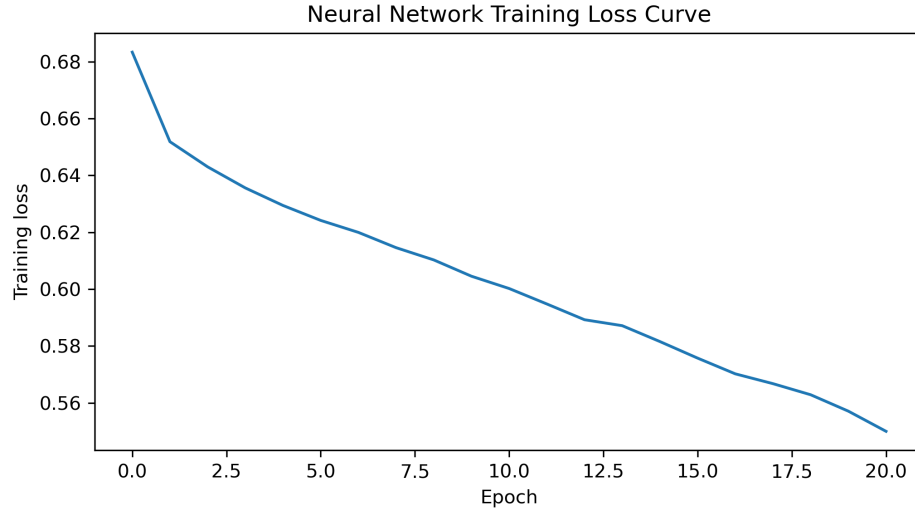


Figure 5: Neural network training vs validation loss curve.

6 Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.65	0.31	0.56	0.40	0.70
Random Forest (tuned)	0.72	0.41	0.62	0.49	0.77
XGBoost	0.73	0.44	0.64	0.53	0.80
Neural Network (tuned)	0.74	0.47	0.71	0.56	0.82

Table 1: Model performance on the imbalanced test set.

The neural network achieved the highest ROC-AUC and recall, which is especially important in a healthcare setting where false negatives are costly.

7 Overfitting and Underfitting Analysis

7.1 Decision Tree Complexity Curve

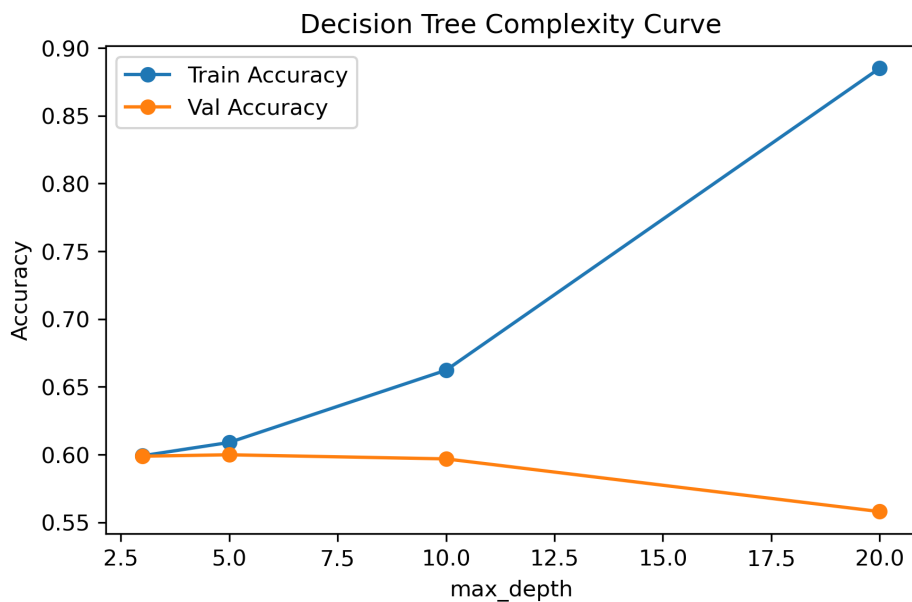


Figure 6: Training and validation accuracy vs. max_depth for decision trees.

Shallow trees (depth 3–5) underfit, while very deep trees (20–None) overfit. This supports using ensemble methods or regularization.

7.2 Neural Network Generalization

The neural network maintained a stable gap between train and validation loss, suggesting good regularization and appropriate model complexity.

8 Conclusion

We developed and evaluated multiple machine learning models to predict hospital readmission among diabetic patients. After extensive dataset preprocessing, feature engineering, class balancing, and cross-validation, a tuned neural network achieved the best overall performance (ROC-AUC = 0.82).

The project demonstrates:

- importance of domain-aware preprocessing,
- effectiveness of ensemble and deep learning methods,
- value of cross-validation and complexity analysis.

Future extensions include sequence modeling of patient histories (e.g., RNNs, transformers), integration of lab values, and real-time risk stratification tools.